R22

01/12/2025

Code No.: 22MEO01

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. (Chem) V Sem (Main/Backlog) Examination December 2025

Principles of Design Thinking

Time: 3 Hours **Max Marks: 60**

Note: "Answer **ALL** questions from **Part-A** and **Part-B** (including **internal choices**) at one place and in the same order as in the question paper."

Part - A

(5Q X 2M = 10 Marks)

			M	CO	BT
1		What is 'Lateral Thinking'?	(2)	1	1
2		Define 'Empathy' in the context of Design Thinking.	(2)	2	1
3		What is meant by 'Problem Statement' in the define phase of design thinking?	(2)	3	1
4		Name any two ideation methods other than Brainstorming.	(2)	4	1
5		List two reasons why prototyping is important.	(2)	5	1

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Explain the impact of engineering on social development.	(5)	1	2
	(b)	Differentiate between the five-stage Design Thinking model and a linear problem-solving approach.	(5)	1	2
		(OR)			
7	(a)	List the different types of thinking and give a brief description of each.	(5)	1	1
	(b)	Explain, with an example, why Design Thinking is considered an iterative process.	(5)	1	2
8	(a)	What are the key outputs of the Empathize phase?	(5)	2	1
	(b)	You are to understand the challenges faced by street vendors. Discuss how you would address the problem faced by street vendors using design thinking.	(5)	2	3
		(OR)			
9	(a)	Describe the steps one should follow to develop empathy towards people.	(5)	2	2
	(b)	Explain the concept of 'Body Storming' with an example.	(5)	2	2
10	(a)	What are the different types of Personas? Briefly explain them.	(5)	3	1
	(b)	For a persona of an "elderly person living alone," create a Point of View (POV) statement.	(5)	3	3
		(OR)			
11	(a)	What is the purpose of 'Why-How Laddering'?	(5)	3	1
	(b)	Analyze how a well-defined problem statement guides the subsequent Ideation phase.	(5)	3	4

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12	(a)	Explain the rules for conducting an effective Brainstorming session.	(5)	4	2
	(b)	Your team is ideating on "how to improve campus security." Apply the Six Thinking Hats method to structure your team's discussion.	(5)	4	3
		(OR)			
13	(a)	What are ideation games? Name any two.	(5)	4	1
	(b)	Compare the use of Mind Maps and Doodling for generating ideas.	(5)	4	4
14	(a)	Develop guidelines for crafting a good Value Proposition.	(5)	5	3
	(b)	A prototype of a new wallet is found to be too bulky by users. Analyze the factors you would consider to make it more desirable.	(5)	5	4
		(OR)			
15	(a)	What is the necessity of the Testing phase?	(5)	5	1
	(b)	Explain the terms 'Desirable', 'Feasible', and 'Viable' in the context of testing a solution.	(5)	5	2

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